

Structuring Clinical Notes into Analyzable Data with R and LLMs

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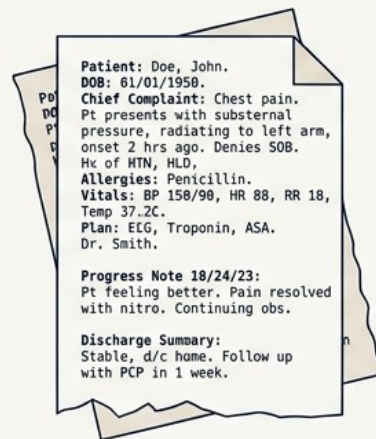
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Last update: Jul 19, 2026

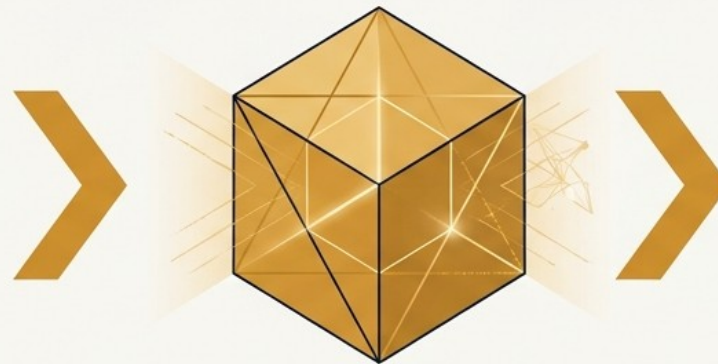
From Text To Structured Data

Transforming unstructured clinical narratives into statistically ready datasets



Unstructured Text

Free-text clinical narratives, progress notes, and discharge summaries.



Large Language Model (LLM)

Acts as the processing engine to parse, interpret, and extract clinical entities.

PATIENT_ID	ENCOUNTER_DATE	CHIEF_COMPLAINT	DIAGNOSIS	MEDICATION	ALLERGY	VITALS_BP	VITALS_HR	PROVIDER
P12345	2023-18-24	Chest pain	Angina HTN, HLD	ASA, Nitro	Penicillin	150/90	88	Dr. Smith
P12345	2023-18-25	Follow up	Resolved	ASA	Penicillin	-	-	Dr. Smith
P12345	2023-18-26	Discharge	Stable	-	Penicillin	-	-	-

Structured Data

Clean, tabular datasets ready for statistical analysis and modeling.

Gemini Notebook's Data Table

Gemini data table example

Clinical Notes Structured Dataset

View prompt and 3 sources

Prompt

Extract clinical notes given in sources into structured dataset for analysis. Main structure: - patient's profile variables - symptoms - signs - physical examinations - laboratory results - imaging findings - diagnosis

Sources

Note 01.docx Note 02.docx Note 20.docx

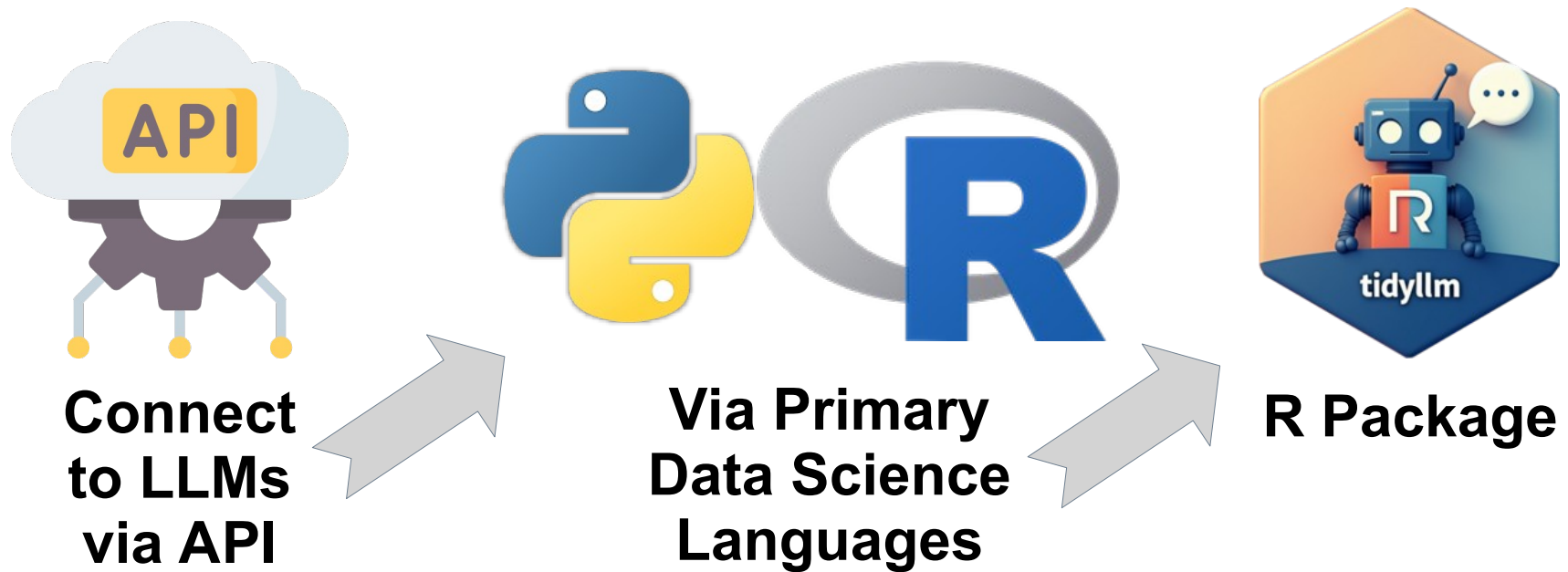
			physical examinations	laboratory results	imaging findings	diagnosis	Source
	le, febrile, multiple bruises over upper and lower limbs,	Palpation of lymph nodes, liver, and spleen	TWBC: $80 \times 10^9/l$, Hb: 7.8 g/dl, Platelet: $20 \times 10^9/l$, blood film: 80% blast cells, bone marrow examination: positive for ALL	Photomicrograph revealed leucocytosis and blast cells	Acute lymphoblastic leukaemia (ALL)	[1]	
	three distinct medical case studies detailing various pathologies across different age groups. The first scenario	bleeding	generalised lymphadenopathy, hepatomegaly (liver 3 cm below right costal margin), splenomegaly (spleen 4 cm below left costal margin)				
Danial, 6-month-old boy	Pale appearance, less active than peers	Pale appearance	Not in source	Genetic tests (parents)	Not in source	Beta thalassemia major [2]	

3 sources →

Good content Bad content

Gemini Notebook can be inaccurate; please double-check its responses.

Programmatic Approach In Research & Practice



Images: flaticon.com, r-project.org, python.org, edubruell.github.io/tidyllm

tidyllm

edubruell.github.io/tidyllm



***“tidyllm** is an R package for working with **large language model APIs** in data analysis workflows. It supports Anthropic Claude, OpenAI, Google Gemini, Mistral, Groq, Perplexity, DeepSeek, OpenRouter, local models via Ollama and llama.cpp, and more — all through a single consistent interface.”*

tidyllm

```
# 1. Define the schema
patient_schema <- tidyllm_schema(
  name = "PatientHistory",
  age = "numeric",
  gender = "character",
  occupation = "character",
  symptoms = "character[]"
)

# 2. Raw unstructured text
raw_text <- "Mrs. Amy, a 35-year-old ward attendant came to the outpatient
            clinic with the complaint of fever for three days and cough."

# 3. Process using a tidy pipeline
full_prompt <- paste("Extract patient note according to the schema.", raw_text)
result <- llm_message(full_prompt) |>
  chat(
    .provider = gemini(), # Gemini as LLM provider
    .model = "gemini-3.1-flash-lite", # Gemini model
    .json_schema = patient_schema # Schema
  )
result

# Structure as JSON
json_result <- get_reply(result) |> minify()
json_result
```

Gemini API

- LLM: Gemini 3.1 Flash Lite
- Get API from Google AI Studio
aistudio.google.com » Projects » + Create a new project
aistudio.google.com » API Keys »  Create API key
- Save API in “api” or “api.txt” in a working folder for our R session

Time for practical session
wnarifin.github.io/workshop